Simulation Study of Free Throw Percentage Estimation Using NBA Data

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Introduction

Statistical estimation studies how well a sample describes an unknown population value. A common example involves repeated events with two outcomes, success or failure. Free-throw shooting in basketball fits this structure. Each attempt results in a made shot or a miss. A player's free-throw percentage represents the probability of success.

Analysts estimate this probability with the sample proportion. The estimator equals the number of made free throws divided by the number of attempts. This estimate changes across samples. Small samples produce large variation. Large samples produce stable estimates. Sample size therefore plays a central role in statistical reliability.

This project examines how sample size affects estimates of a player's free-throw probability. The analysis uses Stephen Curry as the primary example because his free-throw percentage exceeds 0.90 and his career includes thousands of attempts. Data from Basketball Reference provide the number of free throws made and attempted for NBA players. These data allow direct calculation of observed free-throw percentages.

The study treats each free-throw attempt as a Bernoulli trial. Simulation generates repeated samples of free-throw attempts from the estimated probability. These samples form a sampling distribution for the estimator. The analysis studies three statistical properties:

- Variance of the estimator across different sample sizes
- Width of confidence intervals for the free-throw probability
- Coverage rates for 95 percent confidence intervals

Bootstrap resampling also estimates the sampling distribution directly from observed outcomes. This method compares simulated sampling behavior with resampling results drawn from real data. The analysis shows how sample size influences estimator stability and statistical inference in a practical sports context.

Methods

The analysis models each free throw attempt as a Bernoulli trial with two possible outcomes, success or failure. A success represents a made free throw and a failure represents a missed free throw. Let p represent the probability of success for a given player. The estimator of this probability equals the sample proportion of made shots. The estimator takes the form

$$\hat{p} = X / n$$

where X represents the number of successful free throws in a sample of size n.

Statistical theory describes several properties of this estimator. The expected value of the estimator equals the true probability p. This property indicates that the estimator provides an unbiased estimate of the free-throw probability. The variance of the estimator depends on both the probability of success and the sample size. The theoretical variance equals

$$Var(\hat{p}) = p(1 - p) / n$$

This relationship shows how larger samples reduce variability in the estimate. As n increases, the variance decreases, and the estimator becomes more stable.

The analysis studies these theoretical properties using simulation methods. Simulation generates repeated samples of free-throw attempts from a Bernoulli distribution with probability equal to the observed free-throw percentage of a player. Each simulated sample produces an estimate of the free-throw probability. Repeating this process many times produces a sampling distribution of the estimator. This distribution allows direct measurement of estimator variance and sampling behavior across different sample sizes.

The study evaluates several sample sizes. The analysis uses n = 10, 25, 50, 100, and 200 free throw attempts. For each sample size, the simulation generates 1000 repeated samples. The analysis records the sample proportion for each simulation and calculates the variance of the resulting estimates.

The study also evaluates confidence intervals for the free-throw probability. Each simulated sample produces a 95 percent confidence interval for the estimator using the normal approximation. The analysis measures the average width of these intervals and calculates the proportion of intervals that contain the true probability. This measure is known as the coverage rate.

The project also applies a bootstrap resampling procedure. The bootstrap method creates new samples by drawing observations with replacement from the observed sequence of free throw outcomes. Each bootstrap sample produces a new estimate of the free-throw percentage. The distribution of bootstrap estimates approximates the sampling distribution of the estimator based on observed data.

```
simulate_ft <- function(p, n, reps = 1000) {
  makes <- rbinom(reps, n, p)
  estimates <- makes / n
  return(estimates)
}
```

This function generates repeated Bernoulli samples and returns the simulated estimates of the free-throw percentage. The function forms the basis for the simulation analysis used throughout the study.

Simulation

The simulation studies the sampling behavior of the estimator of a player's free-throw probability. The analysis uses NBA player statistics containing the number of free throws made, the number of attempts, and the observed free-throw percentage. The study selects Stephen Curry as the primary example because his free-throw percentage exceeds 0.90 and his career includes a large number of attempts. The observed free-throw percentage serves as the estimated probability of success in the simulation.

The simulation treats each free throw attempt as a Bernoulli trial with probability p equal to the player's observed free throw percentage. For a given sample size n, the number of successful shots follows a binomial distribution. Each simulated sample produces an estimate of the free-throw probability using the sample proportion of successful shots.
The analysis evaluates several sample sizes in order to examine how estimator variability changes as the number of observations increases. The study uses the following sample sizes:

- n = 10
- n = 25
- n = 50
- n = 100
- n = 200

For each sample size, the simulation generates 1000 independent samples. Each sample produces a new estimate of the free-throw percentage. The collection of estimates forms an approximation of the sampling distribution of the estimator. The analysis records the mean of the simulated estimates and the variance of the estimates for each sample size. These results allow direct comparison between the simulated estimator variance and the theoretical variance predicted by statistical theory.

The simulation also evaluates the behavior of confidence intervals for the free-throw probability. Each simulated estimate produces a 95 percent confidence interval using the normal approximation. The analysis calculates the average width of these intervals and measures the proportion of intervals that contain the true probability. This proportion represents the coverage rate of the confidence interval procedure.

The following code demonstrates how the simulation generates repeated estimates of the free-throw percentage and visualizes the resulting sampling distribution.

```
estimates_10 <- simulate_ft(p, n = 10, reps = 1000)
hist(estimates_10,
col = "#4C72B0",
border = "white",
main = "Sampling Distribution of Free Throw Percentage Estimate (n = 10)",
xlab = "Estimated Free Throw Percentage")
```

```
abline(v = p, col = "red", lwd = 2, lty = 2)
```

Repeating this procedure across different sample sizes produces the simulation results analyzed in the following section.

Real Data Analysis

The analysis applies the simulation procedure to NBA free-throw data in order to evaluate the statistical properties of the estimator of a player's free-throw probability. The dataset contains player names, total free throws made, total attempts, and the observed free throw percentage. The study focuses on Stephen Curry because his free-throw percentage is among the highest in the league, and his career includes a large number of attempts. The observed free-throw percentage serves as the estimated probability of success in the simulation model.

Figure 1 shows the sampling distribution of the estimated free-throw percentage when the sample size equals 10 attempts. The distribution displays substantial variability. Individual estimates range widely around the true probability. The red vertical line marks the observed free-throw percentage for the player. Although the distribution centers near the true value, the wide spread indicates large uncertainty when the number of observed attempts remains small.

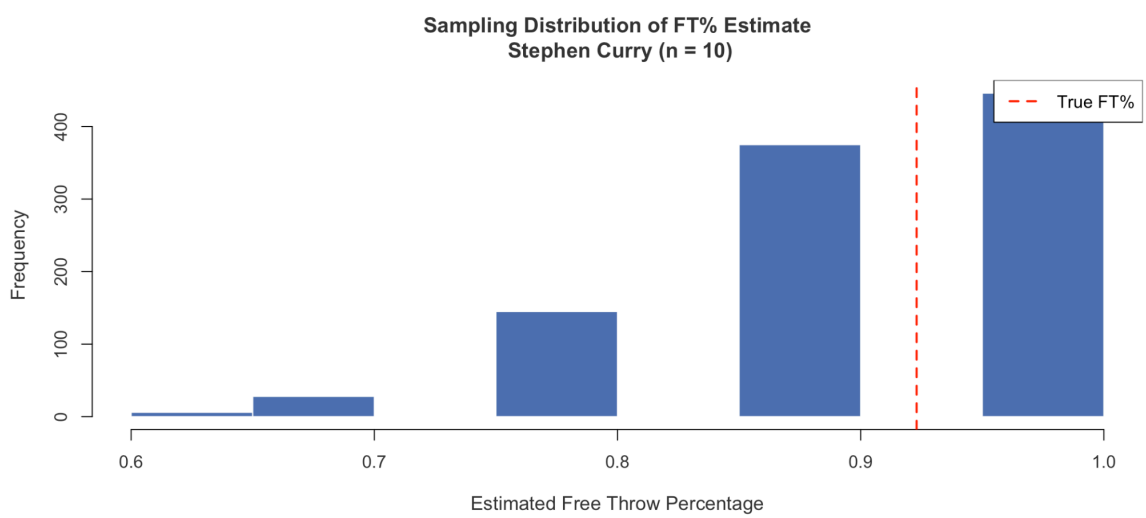


Fig 1. Sampling distribution of the estimated free-throw percentage for n = 10.

Figure 2 presents the sampling distribution when the sample size increases to 100 attempts. The distribution becomes much narrower and concentrates around the true probability. This result demonstrates how larger samples produce more stable estimates. The shape of the distribution also appears more symmetric, which aligns with theoretical expectations for the sampling distribution of the sample proportion.

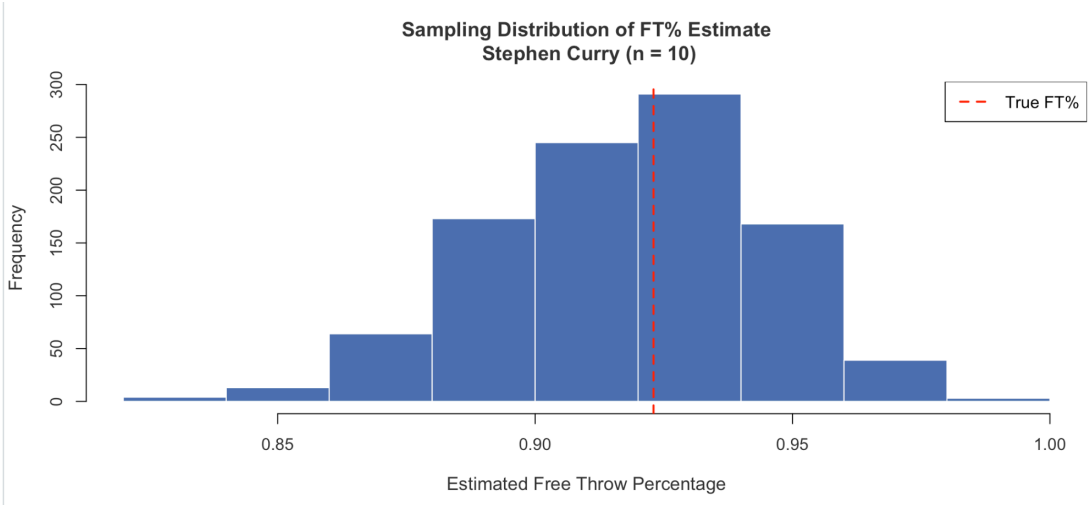


Fig 2. Sampling distribution of the estimated free-throw percentage for n = 100.

The analysis next examines the relationship between sample size and estimator variance. Table 1 reports the mean estimate and simulated variance for each sample size, alongside the theoretical variance predicted by the formula p(1 − p) divided by n.

	SampleSize	MeanEstimate	SimulatedVariance	TheoreticalVariance
1	10	0.926200	0.0065801401	0.007107100
2	25	0.923760	0.0028366991	0.002842840
3	50	0.925260	0.0014934258	0.001421420
4	100	0.922720	0.0006894911	0.000710710
5	200	0.923115	0.0003553772	0.000355355

Table 1. Simulated and theoretical variance of the estimator across sample sizes.

The simulated variance closely matches the theoretical variance predicted by statistical theory. As the sample size increases from 10 to 200 attempts, the variance decreases by nearly an order of magnitude. Figure 3 illustrates this relationship. The simulated variance decreases steadily as the sample size increases. The simulated values follow the theoretical curve predicted by the variance formula.

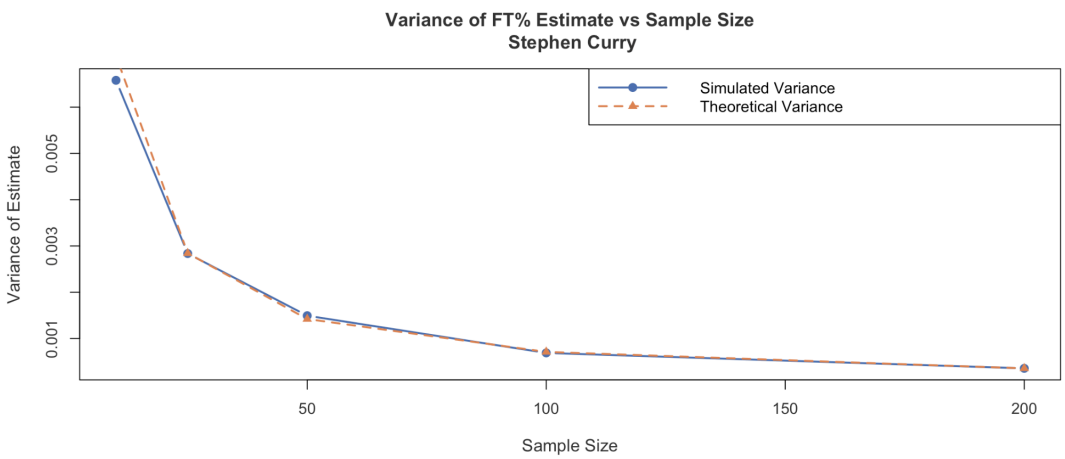


Fig 3. Variance of the free-throw percentage estimator across sample sizes.

The analysis also evaluates the width of 95 percent confidence intervals for the free-throw probability. Table 2 reports the average width of the confidence intervals across the simulation runs. The results show a clear decrease in interval width as sample size increases. Larger samples produce more precise estimates of the probability.

	SampleSize	Avg_CI_Width
1	10	0.18431889
2	25	0.16800702
3	50	0.13760294
4	100	0.10296658
5	200	0.07339329

Table 2. Average confidence interval width across sample sizes.

Figure 4 displays the relationship between sample size and the average width of the confidence intervals. The interval width decreases steadily as the number of attempts increases. This pattern reflects the reduction in estimator variance with larger sample sizes.

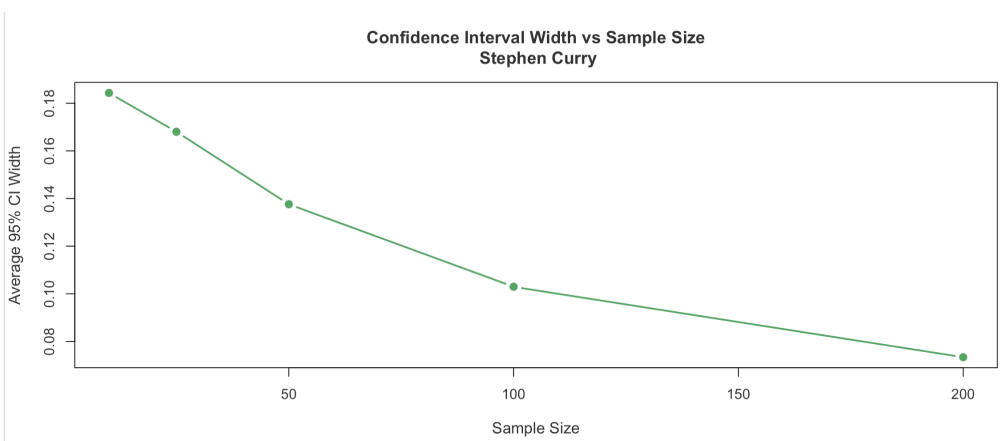


Fig 4. Confidence interval width across sample sizes.

The study also measures the coverage rate of the 95 percent confidence interval procedure. Coverage rate refers to the proportion of simulated intervals that contain the true probability. Table 3 reports the coverage rates for each sample size. Small samples show low coverage rates, while larger samples approach the expected level of 0.95.

	SampleSize	CoverageRate
1	10	0.533
2	25	0.869
3	50	0.894
4	100	0.955
5	200	0.924

Table 3. Coverage rate of 95 percent confidence intervals.

Figure 5 illustrates the relationship between sample size and coverage rate. The results show that small samples produce unreliable intervals, while larger samples produce coverage rates close to the theoretical value.

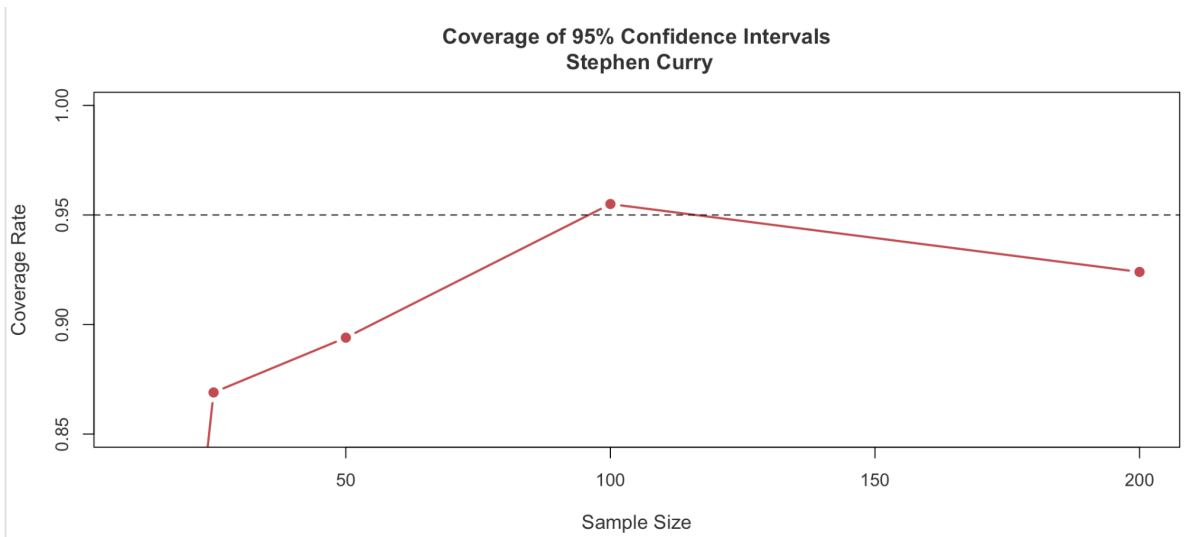
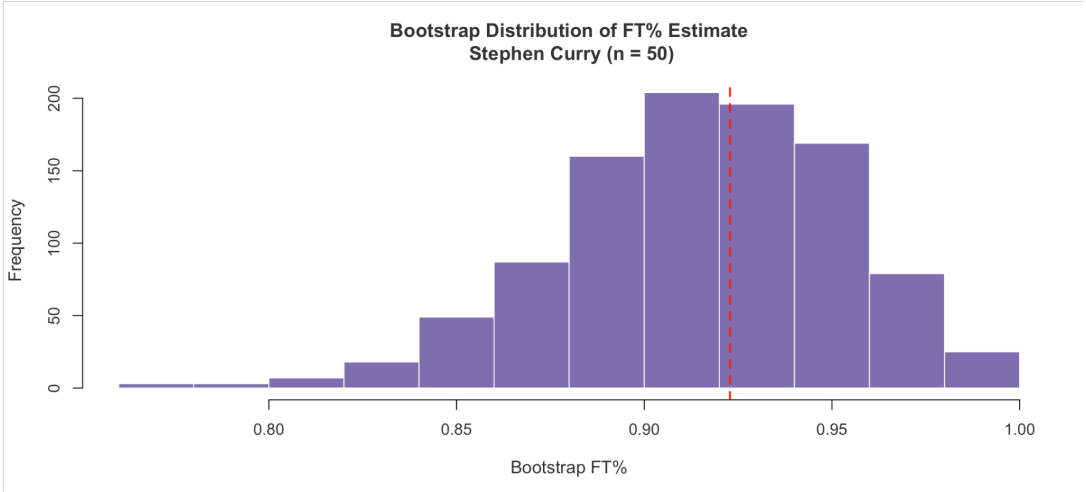


Fig 5. Coverage probability of the 95 percent confidence intervals.

Finally, the study applies bootstrap resampling to estimate the sampling distribution of the estimator directly from the observed free-throw outcomes. The bootstrap method draws samples with replacement from the observed sequence of made and missed shots. Figure 6 displays the resulting bootstrap distribution. The distribution centers near the observed free-throw percentage provide an empirical approximation of the estimator's variability.



Bootstrap distribution of the estimated free-throw percentage.

Conclusion and Discussion

The analysis examined the statistical behavior of the estimator of a basketball player's free-throw probability using simulation and resampling methods. The study treated each free-throw attempt as a Bernoulli trial and used the sample proportion as the estimator of the underlying probability of success. Simulation generated repeated samples across several sample sizes in order to evaluate the variability of the estimator and the performance of confidence interval procedures.

The results show a clear relationship between sample size and estimator stability. Small samples produced large variability in the estimated free-throw percentage. Estimates from samples of ten attempts varied widely around the true value. Larger samples produced much more consistent estimates. When the sample size increased to one hundred or two hundred attempts, the sampling distribution became tightly concentrated around the true probability.

The simulation results also confirmed the theoretical variance formula for the sample proportion. The simulated variance closely matched the predicted value $p(1 - p)$ divided by the sample size. This agreement between simulation and theory supports the use of the binomial model for describing free-throw outcomes.

The analysis of confidence intervals showed that the interval width decreased as the sample size increased. Larger samples produced narrower intervals, which indicates greater precision in the estimate of the free-throw probability. Coverage analysis also demonstrated that the normal approximation for confidence intervals performed poorly for very small samples. Coverage rates approached the expected level of 0.95 only when sample sizes reached one hundred attempts or more.

The bootstrap results provided an additional approximation of the sampling distribution using observed data. The bootstrap distribution centered near the observed free-throw percentage and produced a spread similar to the simulated sampling distribution. This result shows that resampling methods provide a practical way to estimate variability when theoretical assumptions are difficult to verify.

Sports analysts, coaches, and researchers benefit from understanding how sample size affects statistical estimates. Evaluations of player performance based on small samples often lead to misleading conclusions. The results from this study show that larger numbers of observations produce more reliable estimates of shooting ability. This principle applies not only to basketball statistics but also to many other situations in which analysts estimate probabilities from limited data.

References

1. Basketball Reference. NBA Player Statistics. Basketball Reference. https://www.basketball-reference.com/leagues/NBA_2024_totals.html

R Code:

https://github.com/jaxsonkahl/DATA407-FreeThrowProject